

Shyam Patel

Data Scientist

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Summary

- AI/ML Engineer with 3.5+ years of experience designing scalable machine learning and Generative AI solutions. Expert in RAG pipelines, LLM integration, NLP, and cloud-based architectures (AWS/Azure). Proven track record in deploying production-grade ML systems, optimizing performance, and leading AI teams. Strong background in data engineering, Snowflake migration, and Python-based AI-driven automation.
- Beyond technical work, I am passionate about knowledge sharing. I write blogs on Medium, where I break down AI advancements, LLM best practices, and real-world AI applications, making complex concepts accessible to a wider audience

Experience

Kenexai 📍Ahmedabad

Lead AIML Developer (Nov-2025 – Present)

- Designing and developing AI/ML pipelines, RAG systems, and agentic AI solutions.
- Driving AI/GenAI strategy and enterprise architecture design.
- Leading cross-functional AI and data teams.
- Establishing AI governance and deployment standards

Sr. AIML Engineer (Jul-2024 – Oct 2025)

- Architected scalable ML and GenAI solutions (RAG, LLM integration).
- Led production deployments and implemented MLOps practices.
- Mentored junior engineers and optimized performance/cost.

AIML Engineer (Jun-2023 – Jun-2024)

- Built end-to-end ML and Generative AI solutions.
- Developed NLP/LLM pipelines and deployed models on AWS.
- Optimized data workflows and automated processes.

AIML Engineer – Intern (Dec-2022 – May 2023)

- Assisted in developing ML/LLM solutions on AWS.
- Performed data engineering tasks (ETL, data migration, data quality).
- Supported Finance and Home Rental projects using AWS, Oracle, Snowflake, and Informatica Cloud.

MEDIUM Blogger– (Sept-2023 – Present)

- Passionately create and publish insightful, beginner-friendly blogs on Medium.com, aimed at demystifying complex data science concepts for a wider audience.
- Utilize relatable real-life examples to simplify intricate data science topics, attracting an expanding readership and receiving positive feedback from the data science community.

Panth Softech - Software Development Company 📍Ahmedabad:

Data Scientist (Intern) - (May 2022 – July 2022)

- Collaborated with a team of data scientists to develop an innovative image-based deep learning model for automatically extracting temperature panel digits from milk container images.
- Conducted image annotations, participated in model building part.

Skills

Category	Technologies & Tools
Cloud Platforms	AWS (S3, Glue, Lambda, SES, Bedrock, Neptune, RDS, OpenSearch, Sagemaker), Azure Services
Programming	Python (ML Pipelines, AI Models, LLMs, Automation), SQL
Generative AI	RAG, Langchain, LangGraph, LlamaIndex, OpenAI, LLM as Judge, CrewAI, Agents
Automation	N8N, ChatGPT Enterprise
Machine Learning & Deep Learning	ML Algorithms, Neural Network, Transformer, CNN, LSTM
NLP	Named Entity Recognition (NER) models, Text Classification, Text Generation, Metadata Extraction, TTS, STT
Data Platforms & Visualization	Snowflake, Tableau
Version Control	Git, Github
Project Management	JIRA, Confluence
Technical Writing	AI/ML Blog Writing (Medium) on LLMs, AI Advancements, and Best Practices
Soft Skills	Quick Learner, Creativity, Problem Solving

Projects

CallGenie – AI Powered Call Center Analytics System

Technologies Used: Python, OpenAI Whisper, LLM, LangChain, PostgreSQL, PowerBI

Project Description: An AI-driven call analytics system that processes customer-agent conversations to extract insights, track performance, and automate key call center workflows.

Roles and Responsibilities:

- Developed an AI-powered call analytics platform capable of processing and analyzing customer-agent call recordings.
- Integrated speech-to-text technology to generate accurate transcriptions using OpenAI Whisper.
- Implemented sentiment analysis to detect customer emotions, agent tone, and overall call sentiment.
- Designed automated summarization pipelines to produce concise summaries of customer issues and resolutions.
- Built classification modules for identifying issue types (billing, technical, product queries) and detecting resolution status.
- Developed an interactive KPI dashboard showcasing customer satisfaction trends, unresolved cases, and agent performance indicators.
- Enabled a natural language Q&A interface for querying historical call logs (e.g., “Show unresolved billing issues in July”).
- Improved call center efficiency by providing supervisors with actionable insights to enhance agent performance and reduce escalation rates.

AI powered Interview System

Technologies Used: Python, Crew AI, OpenAI, Sql-lite

Project Description: Using AI-driven questioning, resume analysis, and adaptive learning, it enables a fair, scalable, and data-driven hiring process.

Roles and Responsibilities:

- Developed an AI-powered interview bot that analyzes user resumes to extract key skills, technologies, and project descriptions.
- Integrated a large language model (LLM) to understand the context of user-provided resumes and generate interview questions tailored to the applicant's expertise.
- Designed a domain-specific decision-making system where the AI selects appropriate panelists for the interview (e.g., for a Data Engineer, panels include Python, SQL, and Cloud).
- Implemented evaluation steps to assess candidate responses, scoring them on relevance, accuracy, and depth of Understanding.
- Enhanced interactivity by enabling the AI to ask counter-questions on specific topics if clarification or deeper insight is required based on user responses.
- Streamlined the first round of interviews by automating the generation of domain-relevant questions and dynamically adjusting the flow of the interview.
- Improved recruitment efficiency by automating the initial screening process, significantly reducing manual effort.

Custom MCP Integration Platform – Enterprise Process Automation

Technologies Used: Python, MCP (Model Context Protocol), ChatGPT, SharePoint, NetSuite, Oracle Primavera P6 Cloud, REST APIs, PostgreSQL

Project Description: An enterprise AI-powered automation platform that integrates SharePoint, NetSuite, and Oracle Primavera P6 Cloud with ChatGPT through Custom MCP servers, enabling users to interact with enterprise applications using natural language and automate internal business processes.

Roles and Responsibilities:

- Designed and developed Custom MCP servers for integrating enterprise applications with ChatGPT.
- Built secure APIs and connectors for data retrieval, updates, and workflow automation across SharePoint, NetSuite, and P6 Cloud.
- Implemented role-based access control (RBAC), authentication, and audit logging for enterprise security compliance.
- Developed AI-powered business assistants capable of executing actions and retrieving information through natural language interactions.
- Collaborated with business teams to automate manual processes, reducing operational effort and improving productivity.
- Created reusable integration frameworks to accelerate onboarding of additional enterprise systems.

MINT – Media Intelligence & News Tracker System

Technologies Used: Python, GPT-4, Claude, LangChain, FAISS/Pinecone/Milvus, SQL, React, Power BI, NLP

Project Description:

A GenAI-powered media intelligence platform that monitors news and media channels in real time, providing AI-driven summarization, sentiment analysis, risk detection, media query management, and proactive alerting for brand reputation management.

Roles and Responsibilities:

- Developed AI-driven media monitoring and analytics capabilities across Print, Digital, TV, Social, and Vernacular channels.
- Implemented LLM-based summarization, sentiment analysis, entity extraction, topic classification, and risk-scoring pipelines.
- Built semantic search and retrieval systems using vector databases for efficient access to historical media content.
- Developed real-time alerting and dashboard solutions for tracking brand reputation, media trends, and crisis events.
- Created Media Query Management workflows to track journalist interactions and streamline response management.
- Integrated multilingual processing and AI-generated response recommendations to improve communication efficiency

AWS-Powered AI Chatbot

Technologies Used: Python, Langchain, LlamaIndex, Bedrock (Claude), PostgreSQL, AWS (RDS, Neptune, S3, OpenSearch, EC2)

Project Description: Developed an AI-driven chatbot using Retrieval-Augmented Generation (RAG) to retrieve answers from structured and unstructured data sources. The chatbot is built on a full AWS stack, ensuring seamless data retrieval and intelligent response generation.

Roles and Responsibilities:

- Designed and implemented a chatbot leveraging AWS Bedrock (Claude) for LLM-powered natural language interactions.
- Managed structured data in RDS, transforming it into a graph-based model in Neptune for advanced data relationships and querying.
- Constructed SQL queries on Neptune to extract insights from RDS efficiently.
- Handled unstructured data in S3, utilizing OpenSearch for vector storage and intelligent search.
- Deployed a Django-based chatbot application on EC2, integrating all AWS components for a seamless experience.
- Optimized query performance and knowledge retrieval, ensuring context-aware and real-time responses.

Automatic Trip Generation

Technologies Used: Python, AI/ML Algorithms, Maps API, Database Integration

Project Description: Developed a custom logistic system to automate delivery trip planning, optimizing vehicle utilization, reducing delivery time, and minimizing costs through intelligent shipment grouping and route optimization.

Roles and Responsibilities:

- Designed Automated Trip Creation to minimize manual effort by dynamically grouping shipments.
 - Implemented Route Optimization using Maps API for real-time traffic-aware routing.
 - Developed Dynamic Adjustments to integrate new shipments into existing trips with minimal disruption.
 - Built Auto-Adjustment & Merging Mechanisms to optimize vehicle utilization by combining underutilized trips.
 - Enabled Manual Adjustments for fulfillment center users to fine-tune trip allocations.
 - Ensured Scalability by supporting multiple vehicle types and handling fluctuating order volumes.
 - Improved Delivery Timeliness by sequencing shipments efficiently within designated time slots.
- Enhanced Operational Efficiency by reducing errors and automating logistics processes.

AI-Based Email Folder Recommendation System

Technologies Used: Python, Sentence Transformer, Pinecone Vector Database, Named Entity Recognition (NER)

Project Description: Developed an AI-powered system that suggests appropriate folder names for incoming emails using unsupervised learning and similarity-based search, improving email organization efficiency.

Roles and Responsibilities:

- Implemented Text Processing techniques, combining email subject and body while applying preprocessing steps.
- Designed Vector Generation using Sentence Transformer embeddings for efficient text representation.
- Integrated Pinecone Vector Database to store and retrieve high-dimensional embeddings with metadata filtering.
- Applied Named Entity Recognition (NER) for extracting relevant organization-related keywords.
- Developed an Unsupervised Learning Model leveraging similarity search for email classification.
- Achieved 93% accuracy in test cases while ensuring adaptability to new folder structures.

Object Tracking - Flappy Bird Game Using Computer Vision

Technologies Used: Python, OpenCV, Pygame, Computer Vision

Project Description: Developed a Flappy Bird game with an innovative object tracking feature using computer vision techniques. The game utilizes a webcam to track a specific object color, allowing players to control the bird's movement in real time. If the object color is not detected, the game ends.

Roles and Responsibilities:

- Developed the Flappy Bird game mechanics using Python and the Pygame library for GUI development.
- Implemented real-time object tracking using OpenCV, enabling players to control the bird via webcam movement.
- Designed computer vision algorithms to detect and track specific object colors within the video stream.
- Optimized tracking performance to ensure smooth gameplay and accurate object recognition.
- Enhanced user interaction by integrating a dynamic real-time feedback system based on object visibility.

AI-Powered Smart Dashboard Generator and Enhancer

Technologies Used: Tableau, Python, OpenAI API, Pandas, Scikit-learn, LangChain

Project Description: Developed an AI-driven system that automates chart recommendations based on raw datasets and analyzes existing dashboards to suggest improvements for better business insights.

Roles and Responsibilities:

- Implemented genAI models to analyze datasets and recommend the most effective chart types (bar, line, scatter, heatmaps, etc.).
- Developed a reverse analysis engine that evaluates existing dashboards, highlights missing insights, and suggests data-driven enhancements.
- Integrated natural language querying, allowing users to input business questions and receive AI-generated charts with insights.
- Used anomaly detection to identify critical data patterns and suggest visual representations to highlight them.
- Automated KPI tracking by recommending relevant metrics and visuals based on historical data trends.
- Enhanced business decision-making by streamlining data-to-dashboard workflows with AI-powered suggestions.

Container Damage Detection

Technologies Used: OpenCV, Machine Learning, Deep Learning, Flask

Project Description: Developed an AI-powered computer vision system for real-time detection and classification of container damages such as dents, cracks, and corrosion, ensuring cargo safety and operational efficiency.

Roles and Responsibilities:

- Developed a user-friendly interface using Flask for real-time container damage detection.
- Implemented computer vision techniques to detect and classify damages like dents, cracks, and corrosion.
- Built deep learning models using OpenCV to analyze images and identify defects in shipping containers.
- Designed and integrated a system to visualize detected defects and provide detailed location information.
- Optimized the system for real-time analysis to ensure operational efficiency and cargo safety.
- Worked closely with stakeholders to gather requirements and ensure the system met their needs.
- Deployed the system in a real-world environment, enabling real-time detection of container defects.

Speech-to-Text and Text Summarization Project

Technologies Used: Speech Recognition, NLP, Machine Learning

Project Description: Developed an AI-driven speech-to-text and summarization system that converts spoken language into text and generates concise summaries, enhancing content accessibility and productivity.

Roles and Responsibilities:

- Developed speech recognition algorithms to convert spoken language into text, handling challenges like accents and background noise.
- Implemented NLP models to summarize long-form content into concise summaries.
- Integrated speech recognition and NLP technologies to create an end-to-end solution for speech-to-text conversion and summarization.
- Optimized the system to handle different languages and dialects for broader applicability.
- Collaborated with other engineers to fine-tune models and improve the system's overall performance.
- Tested the system with various datasets to ensure robustness and accuracy.

Finance Project - Snowflake Data Migration

Technologies Used: Snowflake, Oracle, SQL, Informatica Cloud, Python

Project Description: Led the migration of financial data from Oracle to Snowflake, enabling a scalable, cloud-based data storage solution. The project involved handling daily and hourly data influx, integrating multiple data sources, and automating data pipelines to ensure seamless migration and efficient processing.

Roles and Responsibilities:

- Migrated financial data from Oracle databases to Snowflake, ensuring data integrity and consistency.
- Collaborated with clients to transition data storage to Snowflake, managing real-time and historical data.
- Worked closely with business users and analysts for requirements gathering and analysis.

- Designed and developed high-level and low-level data architecture based on business needs.
- Implemented Snowflake scripts, stored procedures, and pipelines to automate and optimize data migration.
- Integrated multiple data sources into the staging area for data cleansing and transformation.
- Created and managed Snowflake objects, including databases, schemas, external tables, streams, stored procedures, and tasks, to support ETL workflows.

Achievements and Certifications

NPTEL Certification: Successfully completed the NPTEL Exam with a consolidated score of 72% in [Python for DataScience](#)

Global Representation at GITEX 2024, Dubai: Selected by my company to attend the world's biggest AI event, GITEX 2024 in Dubai, representing the organization and showcasing its innovations.

Coursera Certification: successfully completed the 6 Course Program by Google, which is [Google IT Automation with Python](#).

SnowPro Core Certification Exam (COF-C02): Achieved a score of 700/1000 in the Snowflake Certification Exam.

In addition, I have participated in various hackathons and completed multiple courses on platforms such as Coursera, Great Learning, and Udemy. I also actively solve problems on platforms like HackerRank, LeetCode, and GeeksforGeeks.